Transformation & Plating

M05 | ver 1.1 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Introduce plasmid DNA into TransGen DH5α-type (Trans5α) or TransStbl3 chemically competent E. coli and obtain isolated colonies for screening. Choose conventional heat-shock transformation or the rapid thermocycler route restricted to ampicillin-resistant, high-copy, non-large plasmids.

Materials & Equipment

Item	Manufacturer	Identifier / specification	Use / Specification
DH5α-type cells: Trans5α Chemically Competent Cell	TransGen	CD201	Routine cloning; 50 µL per transformation in the reference setup [1]
TransStbl3 Chemically Competent Cell	TransGen	CD521	Host for constructs prone to unwanted recombination; 50 µL per reference setup [2]
Plasmid DNA or completed assembly reaction	Prepared sample	M04, where applicable	Resistance marker, plasmid size and replication origin known
Sterile LB broth	Solarbio	Liquid bacterial growth medium	Antibiotic-free recovery medium for Route A
LB agar selection plates	As available	Antibiotic matched to plasmid	For Amp selection: 100 µg/mL final ampicillin is a standard working concentration [5]
Control plasmid	Prepared sample	Known compatible plasmid	Positive transformation control
DEPC-treated water	Sangon Biotech	B501005	No-DNA control [6]
Sterile tubes, PCR tubes, tips and spreaders	Axygen tips; laboratory equipment	Compatible sizes	PCR tubes must fit the thermocycler and hold the selected volume
Crushed ice; water bath; shaker; plate incubator	Laboratory ice maker and equipment	42 °C bath; 37 °C incubation	Cooling, conventional heat shock, recovery and colony growth
Thermocycler	Same laboratory thermal cycler as M01	42 °C and 0 °C program	Route B; verify the instrument can reach the specified low temperature

Before You Start

Choose the host and route

Use Trans5α for routine cloning and TransStbl3 for constructs prone to unwanted recombination. Stbl3 host choice does not make a large or unstable construct eligible for Route B. [1, 2]

Route	Choose when	Key sequence
A. Conventional	Default for new assemblies, low-copy or large plasmids, non-Amp selection, or when colony recovery is important	DNA addition → ice incubation → 42°C heat shock → ice → recovery → plating
B. Rapid	Amp resistance AND high copy number AND a non-large plasmid; a familiar, readily transformed construct	DNA addition → 42°C, 2 min → 0°C, 3 min → direct plating

Route B is a laboratory-adapted shortcut; its exact program and efficiency are not established by the cited manuals. Choose Route A if plasmid size, copy number or rapid-route suitability is uncertain. [3]

Prepare DNA, plates and equipment

Per 50 µL cells, start with 1-5 µL containing 1-10 ng purified plasmid, or 1-5 µL completed assembly reaction. Apply any stricter assembly-kit input limit; avoid adding excess reaction mixture. [3]

Warm selection plates to room temperature and remove condensation. For Amp plates, use 100 µg/mL final (1:1000 from 100 mg/mL stock). Prepare antibiotic-free medium and a 42°C water bath for Route A. [5]

For Route B, program 42°C for 2 min, then 0°C for 3 min, once, without a high-temperature lid setting. If the instrument cannot reach 0°C, transfer immediately after heat shock to an ice-water bath for 3 min; record this substitution rather than replacing 0°C with 4°C.

When establishing the method or troubleshooting, include a compatible control plasmid and a no-DNA control. An assembly backbone-only control helps identify cloning background.

Procedure

Common setup

1. Thaw the required competent-cell aliquots on ice. Use 50 µL per transformation as the reference scale and handle gently without vortexing or repeated pipetting. [1, 2]

2. Add the selected DNA input to the cells and mix by gentle tapping. Use a prechilled PCR tube for Route B. Continue with either A or B below, then complete the common plating stage.

Route A. Conventional heat-shock transformation

3. Keep the DNA-cell mixture on ice for 30 min with Trans5α or 20-30 min with TransStbl3. [1, 2]

4. Heat-shock in a 42°C water bath for 45 s, then transfer immediately to ice for 2 min. Keep the tube still during heat shock and cooling. [1, 2]

5. Add 500 µL sterile, antibiotic-free LB. Recover at 37°C, 200 rpm for 1 h, then proceed to plating. [1, 2]

Route B. Rapid transformation for eligible Amp plasmids

6. Immediately after DNA addition, run the thermocycler program below. Omit the prolonged pre-heat-shock ice incubation.

Stage	Temperature	Time
Heat shock	42°C	2 min
Cooling	0°C	3 min

7. After cooling, proceed directly to the Amp plate. Do not add a liquid recovery incubation. Plate the approximately 50 µL cell-DNA mixture, or a smaller portion if dense growth is expected. This route is limited to Amp-resistant, high-copy, non-large plasmids.

Common plating and incubation

8. For Route A, plate 50-100 µL recovered culture on a labeled selection plate; use an additional plate for more culture if colony yield may be low. For Route B, use the direct-plating amount above. Spread evenly with a sterile spreader, avoiding damage to the agar. [3]

9. Allow the liquid to absorb, invert the plates and incubate at 37°C overnight. If the specific construct has a documented lower-temperature propagation requirement, follow that requirement instead and allow sufficient time for colonies to appear. [1, 2]

10. Inspect the sample and control plates together. Select well-isolated colonies for M06 and subsequent construct verification; colony growth alone does not establish the correct insert or sequence.

p.s.

Ampicillin is a bactericidal β-lactam. Omitting liquid recovery for suitable Amp-resistant transformants does not mean Amp is non-killing and should not be generalized to other antibiotics. [3, 4]

Use freshly thawed cells and avoid repeated freeze-thaw cycles. Keep the cell-DNA mixture cold until its selected heat-shock step. [1, 2]

Keep each route's heat-shock device, timing, tube geometry and volume consistent. Do not substitute Route B's 2 min thermocycler step for Route A's 45 s water-bath step alone.

Plate a dilution or use another plate if colonies would be crowded. Avoid prolonged incubation: tiny colonies around larger Amp colonies may be satellites, not independent clones. [5]

Expected Outcome

Expect isolated sample colonies, positive-control growth and little or no no-DNA-control growth. Yield varies with DNA, cells and route; equivalent efficiency is not assumed.

Record the host, route, DNA input, antibiotic and temperature substitutions. Screen and sequence candidate colonies before confirming construct identity.

Troubleshooting

Observed problem	Check first	Action
No colonies, including the positive control	Plate identity and antibiotic activity; thaw history; heat-shock setup	Repeat with fresh cells and a known compatible plasmid using Route A. Check actual bath temperature and timing before changing DNA input.
Few or no sample colonies; positive control grows	DNA quality, assembly success, selection marker and rapid-route eligibility	For a rapid attempt, repeat with Route A. For an assembly, inspect the upstream reaction and its controls; reduce inhibitory reaction carryover rather than simply adding more mixture.
Colonies on the no-DNA control	Plate labeling, missing/inactive antibiotic, or contamination	Test fresh selection plates and repeat the control with clean consumables. Do not interpret the sample colonies as selected transformants until background is resolved.
Confluent growth or spreading patches	Plating density and residual surface liquid	Plate a dilution on a dry plate and allow absorption before inversion. Pick only well-separated colonies.
Tiny colonies around larger Amp colonies	Prolonged incubation and local loss of selection	Treat these as possible satellites. Pick an isolated primary colony; re-streak on fresh selective agar if identity is uncertain.
Colonies grow, but screened constructs are wrong	Backbone-only background, parental plasmid carryover and rearrangement	Review M03/M04 and screen independent colonies. For suspected rearrangement, use the appropriate stable host and construct-specific propagation conditions; increasing colony count alone will not fix the construct.

Reference

[1] TransGen Biotech. Trans5α Chemically Competent Cell, CD201, Version 1.1 (footer V1.1-202512), p. 1: features, procedure and precautions. Accessed 2026-09-19. Open source

[2] TransGen Biotech. TransStbl3 Chemically Competent Cell, CD521, Version 1.1 (footer V1.1-202512), p. 1: features, procedure and precautions. Accessed 2026-09-19. Open source

[3] Addgene. Bacterial Transformation, updated November 13, 2017: DNA input, plating and recovery guidance. Accessed 2026-09-19. Open source

[4] Thermo Fisher Scientific. Ampicillin J66972: General Description and Application (mechanism reference). Accessed 2026-09-19. Open source

[5] Addgene. Pouring LB Agar Plates: antibiotic concentrations and selection notes; last upload September 14, 2016. Accessed 2026-09-19. Open source

[6] Sangon Biotech. B501005 DEPC-treated water, SDS, revision 2024-12-24; product identity. Accessed 2026-10-03. Open source